Plexa Mobility Module: Data Map v3

Supersedes v2. Prepared 14 October 2026. Describes the final state of the NYC DOT traffic volume data after cleaning, combining, storage and the review decisions. Items marked (inference) are conclusions from the data, not from NYC documentation.

Status: data pipeline complete. Parquet archive and PostgreSQL database agree on every count. Baseline design not started.

1. Sources

	Historical	Automated
Raw rows x columns	42,756 x 31 (wide, 24 hourly columns)	1,875,154 x 14 (long, one row per interval)
Date range	2012-01-08 to 2021-05-09	2000-01-01 to 2026-02-03 (bulk 2009 to 2025)
Native resolution	Clock hour	15-minute (most), hourly, odd-minute (quarantined)
Raw SegmentIDs	1,956	3,391
Months with data	Sep to Nov 84%, Jan to May 16%, no Jun to Aug	All twelve months
Geometry	None	WKT point, EPSG:2263

NYC documentation: SegmentID is the LION Segment ID (LION version 14 for the automated dataset). The historical file calls it GIS ID in 2011-12 and Segment ID in 2012-13. Historical ID is a count ID, not a key. Automated RequestID identifies a count request and is not unique per segment.

2. Pipeline and final counts

Stage	Automated	Historical	Combined
Raw rows	1,875,154	42,756	n/a

Stage	Automated	Historical	Combined
Quarantined	19,691 (odd-minute 2,871; EW/NS 6,528; duplicate key 10,292)	350 (duplicate key)	n/a
Hourly rows after zero rule	444,140	1,007,516	n/a
Overlap removed (automated wins)	n/a	22,018	n/a
Rows in combined table	444,140	985,498	1,429,638
Segments	2,843	1,954	4,488

Combined table split for storage: 1,428,980 rows in the hourly table (UTC) and 658 DST-ambiguous rows in hourly_dst_flagged (local time). 1,429,638 in total.

Segments: automated only 2,534; historical only 1,645; both 309. Union 4,488 (2,843 + 1,954 - 309).

Road types in combined table: unspecified 1,424,550; service 3,024; main 2,064.

Identity conflicts (parquet count, including DST rows): 11,849. In the database (non-DST): 11,844. Baseline-eligible rows: 1,412,050 in both.

Volume check: automated database sum equals parquet sum; historical database sum equals parquet sum (non-DST rows, exact).

3. Cleaning rules as applied

Automated

- Vol: commas removed; one negative value (-1) set to missing.
- Timestamp: built from Yr, M, D, HH, MM; interval start convention confirmed.
- Study type per RequestID, SegmentID and Direction: hourly if all MM values are 0, otherwise 15-minute.
- Hour build: 15-minute studies need all four intervals; hourly studies need one value.
- Quarantine priority: odd-minute, then EW/NS, then duplicate key (same SegmentID, Direction, timestamp).

- Zero rule (both sources): a zero run of 2 hours or more is set to missing. A shorter run is set to missing when run length times the segment's typical hourly volume is at least 8 vehicles. Otherwise the zero is kept. Automated: 67,965 + 9,822 zero cells set to missing, 17,491 kept.
- Fragments: 548 segments (18,950 raw rows) have no complete hour. Saved separately.

Historical

- Direction split into base direction (NB, SB, EB, WB) and road type (unspecified, main, service).
- Quarantine: duplicate key on (SegmentID, direction, road type, date): 350 rows.
- Zero rule as above: 6,947 + 249 zero cells set to missing; 373 kept (typical volume at most 7.5 per hour).
- Result: 1,007,516 hourly rows over 1,954 segments. The two dropped raw segments (46664, 9001153) had all rows in quarantine as duplicate keys.

Local time: hours are stored as local clock time with a timezone-aware copy. DST-ambiguous hours (105 automated, 553 historical after overlap removal) carry dst_issue = True.

4. Segment identity and LION join

LION (DCP, EPSG:2263) provides geometry and road attributes. Alternate names come from the altnames layer, joined on Join_ID (100 percent match). The centerline (CSCL) file does not match SegmentIDs; its PhysicalID and OBJECTID overlap by chance and are not used.

Match order per segment:

1. LION SegmentID, if the count point lies within 100 feet of the matched line.
2. LegacyID, if within 100 feet and the street name agrees (fuzzy, abbreviation-aware, altnames included).
3. Nearest line, automated only, if the street name agrees: 111 segments.
4. None: attributes left empty.

Automated: SegmentID 2,892; LegacyID 335; spatial 111; none 53. No segment matched two different LION rows.

Historical: automated_validated 345; SegmentID 1,406; LegacyID 165; none 38.

Segment table (4,488): lion_key_used = SegmentID 3,801; LegacyID 446; spatial 94; none 84; automated_validated 37; identity_conflict 26.

LION attributes: posted speed, travel lanes, park lanes, total lanes, street width min and max, segment length, traffic direction, truck route type, snow priority, bike lane, LBoro, RBoro. StreetWidth_Irr and FCC are blank in LION and dropped. Blanks in park lanes, truck route and bike lane are recoded to 0 for matched segments (inference: blank means none). Posted speed is 25 mph for 80 percent of matched rows.

5. Identity conflicts and review decisions

A historical row is an identity conflict when its recorded street does not agree with the matched LION street after normalisation and alias matching. Conflict rows keep their volume, have LION attributes removed, carry identity_conflict = True, and are excluded from baselines.

Review of conflict segments (after alias and altnames pass):

- Segment 82920 (Van Wyck Expressway Exit 11 SB, 2014): false positive. Automated and historical names agree; 48 combined rows reinstated.
- Segment 29555 (Fulton Avenue, 2012): false positive. Automated and historical names agree; 251 rows reinstated.
- Segments 110463 (Grand Ave vs Grand St), 23207 (Liberty Ave vs Liberty St), 95270 (Lexington Ave vs Hutchinson River Pkwy), 4340 (4th Ave vs Arthur Kill Rd): real conflicts; stay excluded.
- 42 conflict segments have no automated record and cannot be checked by name. They stay excluded. This is the main residual exposure. Their conflict is supported only by LION and its alternate names.

After review, 46 historical segments carry conflict rows. The conflict review table is saved as conflict_review.csv.

6. Borough

Rule: borough is the automated recorded Boro. Where automated is missing, LION borough (LBoro code 1 to 5) is used. Manual or reviewed decisions override the rule.

Sources in segment table: automated 2,840; LION 1,578; manual override 3 (Roosevelt Island Main St: 36764, 36905, 114477, Manhattan); reviewed 3; automated_unresolved 1; none 67 (historical-only, no LION match, left empty).

Reviewed decisions (16 segments disagree between LION and automated; boro_disagreement = True for all 16):

- 155079 Queensboro Bridge upper roadway: Manhattan (reviewed; medium-high).
- 158339 Pulaski Bridge: Brooklyn (reviewed; medium-high).
- 66420 Grand St Bridge: Brooklyn (reviewed; medium, border bridge).
- 136594 Nostrand Ave at Vernon Ave: kept Brooklyn, unresolved (automated_unresolved). The LION value is Queens. Evidence is weak either way.
- Other 12: automated value kept, based on street names and neighbour votes.

LION and automated disagreements are not boundary effects: LBoro equals RBoro for all 16.

7. Seasonality and coverage (baseline rows)

Season share of baseline rows: September to November 67 percent; December to February 13 percent; March to May 13 percent; June to August 7 percent.

Segment-directions carrying baseline data: 5,899. 4,917 (83 percent) observed in one season; 900 in two; 82 in three; none in all four.

Summer to autumn comparison, road-years observed in both seasons of the same year, with at least 3 full days in each season: 17 road-years in 6 years. Median ratio about 1.02 (full-days method 1.015). Interval about 0.90 to 1.10. At a 5-day threshold only 5 road-years remain, which is not reportable. Three 2019 outliers were single-day autumn artifacts and are excluded.

Conclusion: no detectable summer difference from autumn on these roads, within about plus or minus 10 percent. No summer multiplier is applied. No seasonal curve is estimable.

Winter to summer: zero road-years observed in both in the same year. Not identifiable.

COVID window: no baseline rows for April to July 2020 in either source. Other 2020 rows are August (automated) and October to November (historical). The covid flag from 2020-03-01 describes the post-spring period, not a measured lockdown.

Topic summer coverage: all summer data comes from automated. Historical has no June to August data.

8. Storage

PostgreSQL 16 with TimescaleDB and PostGIS, in Docker: container plexa-db, database plexa, port 5433. Setup script setup_plexa.sql drops and recreates the tables; the loader load_plexa.py follows it.

Tables:

- segments (4,488 rows): segment_id primary key; borough, borough_source, boro_disagreement; LION key and attributes; geometry lion_geom (MultiLineString, EPSG:2263) and geom_4326 (Point); coverage flags and time spans; conflict counts.
- hourly (hypertable, 1,428,980 rows): primary key (segment_id, direction, road_type, hour_start_utc). Monthly chunks. Partial index on baseline_ok rows. Columns include volume, source, study_id, identity_conflict, partial_day, covid, sparse_year, baseline_ok.
- hourly_dst_flagged (658 rows): same measures, keyed by local clock time, because the ambiguous hour cannot be converted to UTC without guessing which pass it belongs to.
- Quarantine and fragments: CSV files, not in the database.

Load checks passed: segments 4,488; hourly 1,428,980; DST 658; baseline 1,412,050 after the 29555 fix; no orphan hours; LION geometry present for every segment with a LION match; hypertable registered; volume sums match the parquet files exactly for both sources.

Backup: the container has no data volume. pg_dump before any container removal.

9. DST rule

DST-ambiguous hours are kept with the dst_issue flag and excluded from baselines. Their share is about 0.05 percent of rows. Fall-back days show the repeated 1 AM hour only once, so its volume may cover one pass, both, or a mix. The rule keeps the rows available and removes the ambiguity from the hour-of-day profile.

10. Limitations for modelling

- Seasonal: no seasonal curve; summer uncertainty about plus or minus 10 percent; no summer multiplier.

- Coverage: most roads are observed in one season and often one campaign. Per-road seasonality cannot be learned for most segments.
- Year concentration: 2012 holds 210,188 combined rows, mostly historical autumn campaigns. Years 2000 to 2008 and 2026 are sparse, and 2013, 2018 and 2021 are thin.
- Borough balance: no Bronx roads appear in the same-year summer comparison; Brooklyn and Queens dominate.
- Identity conflicts: 0.85 percent of combined rows excluded, including 42 unverifiable segments.
- Zero handling: the zero rule is a documented choice, not validated against ground truth.
- LION version: automated documentation cites LION version 14; the LION file used is newer, so some IDs may have been reassigned. LegacyID and the 100-foot test are the safeguards. Name agreement on matched segments is 84 to 94 percent.
- Partial days: 18 percent of automated and 2 percent of historical hourly rows fall on days with fewer than 24 valid hours. Flagged with partial_day.
- Daylight saving: see section 9.
- Segment table conflict counts (hist_rows, conflict_rows) are raw pre-overlap counts. Baselines use the hourly table, not these counts.

11. Decision log (v2 to v3)

Topic	v2	v3	Reason
Segment 82920	Excluded, identity conflict	Reinstated (48 rows)	Same ramp name and year in both sources
Segment 29555	Excluded, identity conflict	Reinstated (251 rows)	Same street name in both sources
Other checkable conflicts	Excluded	Excluded (4 segments, real conflicts)	Names differ
Unverifiable conflicts	Excluded	Excluded (42 segments)	No automated record; flagged as limitation
DST hours	In baseline	Excluded from baseline, kept in table	Ambiguous volume

Topic	v2	v3	Reason
Borough	Automated, LION fallback, 16 flagged	Same rule plus 3 reviewed, 1 unresolved	Review of street names and neighbours
EW and NS	Quarantined, documented as two-way totals	Quarantined; corridor test found no paired corridor-hours	Documentation not verified either way; excluded from one-way baseline in any case
Storage	Not started	PostgreSQL with TimescaleDB and PostGIS; DST rows in a separate table	Counts reconcile

12. Open items

1. Odd-minute studies and EW/NS: meaning undocumented. Quarantined. Could be resolved by a location test (NB and SB count points within about 50 feet in the same hour), if needed for anything else.
2. Segment 136594: borough unresolved.
3. 42 conflict segments without an automated record: exclusion is a limitation; no further action unless a historical name check becomes available.
4. Retire data map v2, hist_hourly_v1, and earlier automated versions (v1 to v3).
5. Correct the earlier '597 studies (2.4 percent)' statement in v2: it is unsupported. Clean count of rows from hourly studies is 10,983.
6. Back up the database before any container removal.
7. Baseline design: road-level profile by hour and day of week, using baseline_ok rows, with season recorded and summer uncertainty stated. Not started.
8. Evaluation: within-season holdout on the same roads; segment holdout for unseen roads; season-transfer test on roads observed in both seasons. Not started.
9. Experiment tracking with MLflow, starting from the historical average by segment, hour and day of week. Not started.

13. Files

Parquet (data/clean):

- auto_hourly_v4.parquet: automated hourly, 444,140 rows
- hist_hourly_v2.parquet: historical hourly, 1,007,516 rows
- combined_hourly_v3.parquet: combined hourly, 1,429,638 rows, 17 columns (current)
- segment_table_v4.parquet: segments, 4,488 rows (current)
- auto_quarantine.csv.gz, auto_fragments.csv.gz, hist_quarantine.csv.gz
- hist_identity_conflict_groups.csv, conflict_review.csv, borough_decisions.csv

Superseded: auto_hourly_v1 to v3, hist_hourly_v1, combined_hourly_v1 and v2, segment_table_v1 to v3.

Database: plexa (Docker container plexa-db, port 5433) with segments, hourly, hourly_dst_flagged.

14. Column dictionary, combined_hourly_v3

Column	Meaning
segment_id	LION Segment ID
direction	NB, SB, EB or WB
road_type	unspecified, main or service
hour_start	Start of the clock hour, local time, naive
hour_start_tz	Same hour, timezone-aware (America/New_York); null on DST-ambiguous hours
volume	Vehicles in the hour
source	automated or historical
study_id	RequestID (automated) or count ID (historical); unique only within source
orig_interval_min	15 or 60 (study type)
dst_issue	True on DST-ambiguous hours

Column	Meaning
identity_conflict	True for historical rows whose recorded street disagrees with LION
date	Calendar date of the hour
hours_in_day	Valid hours for this segment, direction, road type and date
partial_day	True when hours_in_day is below 24
covid	True from 2020-03-01
sparse_year	True for 2000, 2006, 2007, 2008, 2013, 2018, 2021, 2026
baseline_ok	True when road_type is unspecified, identity_conflict is False, and dst_issue is False